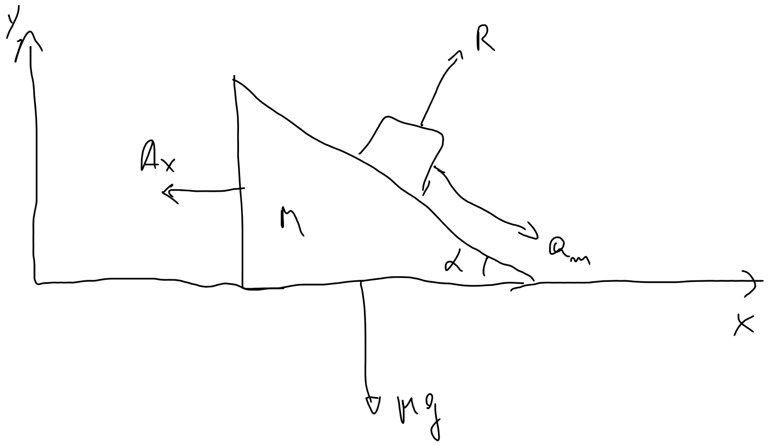


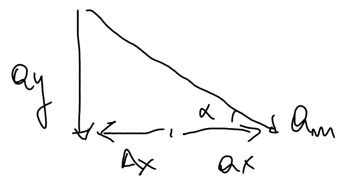
$$\begin{cases} a_x = A_x + a_m \cos\alpha \\ a_y = -a_m \sin\alpha \end{cases}$$

$$a_y = (A_x - a_x)\,\text{tg}\,\alpha$$

$$\Leftrightarrow \text{tg}\,\alpha = \frac{a_y}{A_x - a_x}$$

$$y(t) = a_y \frac{t^2}{2}$$

$$X(t) = A_x \frac{t^2}{2}$$



$$a_m = -\frac{a_y}{\sin\alpha} = \frac{a_x - A_x}{\cos\alpha}$$

$$\begin{cases} \sin\alpha = -\frac{a_y}{a_m} \\ \cos\alpha = \frac{a_x - A_x}{a_m} \end{cases}$$

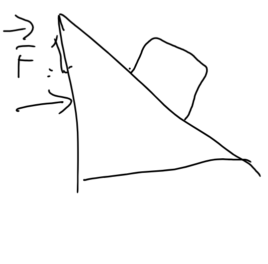
Se $A_x = a_x \Rightarrow a_y = 0$

$$a_x = \frac{Mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha}$$

$$A_x = \frac{-Mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha}$$

$$a_y = (A_x - a_x)\,\text{tg}\,\alpha = \frac{-(M+m)\,g\sin^2\alpha}{M + m\sin^2\alpha}$$

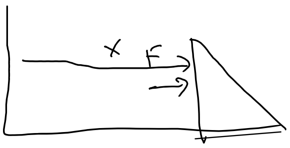
Se $A_x = a_x \Rightarrow \frac{Mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha} = -\frac{m\,g\sin\alpha\cos\alpha}{M + m\sin^2\alpha} \Rightarrow \boxed{m = -M}$



$$F - R\sin\alpha = M A_x \Rightarrow A_x = \frac{F}{M} - \frac{m a_x}{M}$$

$$R\sin\alpha = m a_x \Rightarrow R = \frac{m a_x}{\sin\alpha}$$

$$R\cos\alpha - mg = m a_y$$



$$x = X + s\cos\alpha \Rightarrow a_x = A_x + a_m\cos\alpha$$

$$a_y = -a_m\sin\alpha \qquad a_m = -\frac{a_y}{\sin\alpha} = \frac{a_x - A_x}{\cos\alpha}$$

$$a_y = (A_x - a_x)\,\mathrm{tg}\,\alpha$$

$$a_y = \frac{R}{m}\cos\alpha - g = a_x\,\mathrm{cotg}\,\alpha - g$$

$$a_x\,\mathrm{cotg}\,\alpha - g = \left(\frac{F}{M} - \frac{m a_x}{M} - a_x\right)\mathrm{tg}\,\alpha = -\frac{(M+m)}{M}a_x\,\mathrm{tg}\,\alpha + \frac{F}{M}\,\mathrm{tg}\,\alpha$$

$$a_x\,\mathrm{cotg}\,\alpha + \frac{(M+m)}{M}a_x\,\mathrm{tg}\,\alpha = g + \frac{F}{M}\,\mathrm{tg}\,\alpha$$

$$(M\cos^2\alpha + M\sin^2\alpha + m\sin^2\alpha)\,a_x = Mg\sin\alpha\cos\alpha + F\sin^2\alpha$$

$$a_x = \frac{Mg\sin\alpha\cos\alpha + F\sin^2\alpha}{M + m\sin^2\alpha} = a_x' + \frac{F\sin^2\alpha}{M + m\sin^2\alpha}$$

$$A_x = \frac{F}{M} - \frac{m}{M}a_x = \frac{F}{M} - \frac{m}{M}a_x' - \frac{m}{M}\frac{F\sin^2\alpha}{M + m\sin^2\alpha}$$

$$A_x = -\frac{m}{M}a_x' + \frac{F}{M}\left(\frac{M + m\sin^2\alpha - m\sin^2\alpha}{M + m\sin^2\alpha}\right)$$

$$A_x = \frac{-mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha} + \frac{F}{M + m\sin^2\alpha}$$

$$a_y = (A_x - a_x)\,\mathrm{tg}\,\alpha =$$

$$A_x - a_x = \frac{F(1 - \sin^2\alpha)}{M + m\sin^2\alpha} - \frac{(m+M)g\sin\alpha\cos\alpha}{M + m\sin^2\alpha}$$

$$a_y = \frac{F\cos^2\alpha\,\mathrm{tg}\,\alpha}{M + m\sin^2\alpha} - \frac{(M+m)g\sin\alpha\cos\alpha\,\mathrm{tg}\,\alpha}{M + m\sin^2\alpha}$$

$$a_y = \frac{F\sin\alpha\cos\alpha}{M + m\sin^2\alpha} - \frac{(M+m)g\sin^2\alpha}{M + m\sin^2\alpha}$$

$$a_x = a_x' + \frac{F\sin^2\alpha}{M+m\sin^2\alpha}$$

$$\operatorname{tg}\alpha = \frac{a_y'}{A_x' - a_x'} = \frac{a_y}{A_x - a_x}$$

$$A_x = A_x' + \frac{F}{M+m\sin^2\alpha}$$

$$A_x - a_x = A_x' - a_x' + a_R$$

$$a_y = a_y' + \frac{F\sin\alpha\cos\alpha}{M+m\sin^2\alpha}$$

$$a_R = \frac{F}{M'} - \frac{F\sin^2\alpha}{M'}$$

$$a_R = \frac{F\cos^2\alpha}{M'}$$